

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Designed agentic AI workflows using OpenAI Agents SDK and MCP tool-calling; built RAG retrieval layer with Pinecone vector store handling millions of daily transactions at 99%+ uptime.
- Built a deployment-gated evaluation pipeline running automated quality, latency, and drift checks before every release, blocking 15% of releases that failed validation criteria before reaching production.
- Architected AWS serverless infrastructure (Lambda, S3, DynamoDB, API Gateway, SageMaker) enabling AI agents to interact with enterprise systems via production APIs at sub-200ms latency and scale.

QureAI

Jan 2024 – May 2025

ML Infrastructure Engineer

- Improved RAG retrieval recall 61% to 89% on a 2K-query eval set by rebuilding chunking pipeline with hybrid dense-sparse retrieval and re-ranking.
- Cut p99 inference latency 420ms to 140ms across 50K+ daily requests migrating synchronous serving to async vLLM on GCP Cloud Run with autoscaling.

Nomura Research Institute Financial Technologies

Aug 2020 – Aug 2023

Software Engineer

- Built and operated 25+ production microservices in C++ and Java-backed distributed systems for 1M+ users, sustaining 99.9% uptime over 3 years on AWS (EC2, RDS, SQS).
- Cut nightly financial data aggregation blocking downstream settlement from 8 minutes to 90 seconds by redesigning SQL with indexed views and batch partitioning.
- Standardized failure-handling contracts across 25+ financial services, reducing production escalations 60% in one quarter via consistent retry, timeout, and propagation semantics.

Projects

RootCause AI | github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed root cause reports.

LangFetch | tinyurl.com/LangFetch

- LangGraph agent with Pinecone vector retrieval and hybrid re-ranking decomposes multi-table questions into dependency-ordered SQL subqueries; deployable as MCP-compatible server for enterprise data queries.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, C++, Java, TypeScript, SQL

Agentic AI: LangGraph, MCP, OpenAI Agents SDK, Claude, LangChain, RAG, Vector DBs (Pinecone, FAISS), Prompt Engineering, vLLM

Backend & APIs: FastAPI, REST APIs, Microservices, PostgreSQL

Cloud & Ops: AWS (Lambda, EC2, S3, RDS, SQS, SageMaker), GCP (Cloud Run), Docker, Kubernetes, CI/CD

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer